# Mth 603 MCQS

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Q#1

While solving the system; x+4y=4, 2x-y=1 by Gauss-Seidel method, which of the following ordering will lead more rapidly to the desired order of accuracy?

**No need to rendering**

x-2y=1, x+4y=4

x+4y=1, 2x-y=4

2x+y=1, x+4y=4

Q#2

Sparse matrix is a matrix with ……….

Some elements are zero

**Many elements are zero**

Some elements are one

Many elements are one

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Q#3

While solving a system of linear equations, which of the following approach is economical for the computer memory?

Direct

**Iterative**

Analytical Graphical

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Q#4

**Numerical methods for finding the solution of the system of equations are classified as direct and ………… methods**

**Indirect**
**<u>Iterative</u>**

**Jacobi**

**None of the given choices**

**Page # 48**

Q# 5
Back substitution procedure is used in **…………….**
**<u>Gaussian Elimination Method</u>**

Jacobi's method

Gauss-Seidel method

None of the given choices

Q# 6

If the determinant of a matrix A is not equal to zero then the system of equations will have……….
**<u>Unique solution</u>**

Many solutions

Infinite many solutions

None of the given choices

Q#7
*In Jacobi's Method, we assume that the ______ element does not vanish.*
***<u>Diagonal</u>***
*Off diagonal*
*Row*
*Column*

Q#8

The linear equation: x+y=1 has --------- solution/solutions.
***<u>Unique</u>***

*No solution*
*Infinite many*
*Finite many*

Q#9

While using Relaxation method, which of the following is the largest Residual for 1st iteration on the system; 2x+3y = 1, 3x +2y = - 4 ?

-4
**3**

**2**

1

Q#10

If the determinant of a matrix A is equal to zero then the system of equations will have……….

Unique solution

No solutions

Infinite many solutions
**No Solution or Infinite many solutions**

2

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Question # 1 of 10 ( Start time: 10:15:15 PM ) Total Marks: 1

If the pivot element happens to be zero, then the i-th column elements are searched for the numerically …………… element.

Select correct option:

Smallest

**Largest**

Question # 2 of 10 ( Start time: 10:15:30 PM ) Total Marks: 1

In Jacobi's Method, We assume that the …………elements does not vanish.

Select correct option:

**Diagonal**
Off-diagonal

Row

Column

Question # 3 of 10 ( Start time: 10:16:48 PM ) Total Marks: 1

Which of the following method is not an iterative?

Select correct option:

Gauss–Seidel method

Iteration's method

Relaxation Method

**Gauss Jordan method**

Question # 4 of 10 ( Start time: 10:17:25 PM ) Total Marks: 1

While using Relaxation method, which of the following is increment 'dxi'corresponding to the largest Residual for 1st iteration on the system; 2x+3y = 1, 3x +2y = - 4 ?

Select correct option:

-2

2

**3**

4

Question # 5 of 10 ( Start time: 10:18:33 PM ) Total Marks: 1

While solving a system of linear equations, which of the following approach is economical for the computer memory?

Select correct option:

Direct

**Iterative**

Analytical

Graphical

Question # 6 of 10 ( Start time: 10:19:08 PM ) Total Marks: 1

Jacobi's Method is a/an………………

Select correct option:

**Iterative method**

Direct method

Question # 7 of 10 ( Start time: 10:19:51 PM ) Total Marks: 1

Numerical methods for finding the solution of the system of equations are classified as direct and ………… methods

Select correct option:

Indirect

**Iterative**

Jacobi

None of the given choices

Question # 8 of 10 ( Start time: 10:20:19 PM ) Total Marks: 1

For a system of linear equations, the corresponding coefficient matrix has the value of determinant; $|A| = -3$, then which of the following is true?

Select correct option:

The system has unique solution

**The system has finite multiple solutions not sure**

The system has infinite may solutions

The system has no solution not sure

Question # 9 of 10 ( Start time: 10:21:36 PM ) Total Marks: 1

While solving the system of linear equations; x-y=2, -x+y=3 by Jacobi's method, if (0,0) be its first approximate solution, then which of the following is second approximate solution?

Select correct option:

**(2,3)**

(2,0)

(0,3)

Second approximate solution doesn't exist as system is singular

Question # 10 of 10 ( Start time: 10:23:02 PM ) Total Marks: 1

In …………… method, a system is reduced to an equivalent diagonal form using elementary transformations.

Select correct option:

Jacobi's

Gauss-Seidel

Relaxation

**Gaussian elimination**

Question # 1 of 10 ( Start time: 10:47:21 PM ) Total Marks: 1

Full pivoting, in fact, is more ……………than the partial pivoting.

Select correct option:

Easiest

**Complicated**

Question # 2 of 10 ( Start time: 10:48:21 PM ) Total Marks: 1

Which of the following is the meaning of partial pivoting while employing the row transformations?

Select correct option:

**Making the largest element as pivot**
Making the smallest element as pivot

Making any element as pivot

Making zero elements as pivot

Question # 3 of 10 ( Start time: 10:49:09 PM ) Total Marks: 1

The basic idea of relaxation method is to reduce the largest residual to …………

Select correct option:

One

Two

**Zero**
None of the given choices

Question # 4 of 10 ( Start time: 10:49:34 PM ) Total Marks: 1

Which of the following method is not an iterative method?

Select correct option:

Jacobi's method

Gauss-Seidel method

Relaxation methods

**Gauss-Jordan elimination method**

Question # 5 of 10 ( Start time: 10:50:16 PM ) Total Marks: 1

The system of linear equations; 2x+3y = 6, x+4y = 5 is not strictly diagonal dominant as---------

Select correct option:

|2| < |4|

|2| < |1|

|3| < |4|

|2| < |3|

Question # 6 of 10 ( Start time: 10:51:42 PM ) Total Marks: 1

If the pivot element happens to be zero, then the i-th column elements are searched for the numerically …………… element.

Select correct option:

Smallest

**Largest**

Question # 8 of 10 ( Start time: 10:52:43 PM ) Total Marks: 1

While solving a system of three linear equations in three variables by Jacobi's method, the first approximate solution is NECESSARY to be taken as (0,0,0).

Select correct option:

True

**False**

Question # 9 of 10 ( Start time: 10:54:07 PM ) Total Marks: 1

If the Relaxation method is applied on the system; 2x+3y = 1, 3x +2y = - 4, then largest residual in 1st iteration will reduce to ---------.

Select correct option:

**zero**

4

-1

-1

Gauss elimination and Gauss-Jordan methods are popular among many methods for finding the ………of a matrix.

Select correct option:

**Inverse**

While solving the system; x–2y = 1, x+4y = 4 by Gauss-Seidel method, which of the following ordering is feasible to have good approximate solution?

Select correct option:

**no need to reordering**

While using Relaxation method, which of the following is the Residuals for 1st iteration on the system; 2x+3y = 1, 3x +2y =4 ?

**(1,4)**

Back substitution procedure is used in …………….

**Gaussian elimination method**

For two matrices A and B, such that "A = Inverse of B", then which of the following is true?

b= inversea

a and b ARE NON LINEAR

**AB= identity matrix**
All choice true

Sparse matrices arise in computing the numerical solution of …………….

Select correct option:

Ordinary differential equations

**Partial differential equations (Page 69)**
Linear differential equations

Non-linear differential equations

Which of the following is equivalent form of the system of equations in matrix form; AX=B ?

Select correct option:

XA = B

**X = B(Inverse of A)**
X =(Inverse of A)B

BX = A

In …………… method, a system is reduced to an equivalent diagonal form using elementary transformations.

Select correct option:

JCB

GUASS

RELAXTION

**GUASS Elimination**

Gauss–Seidel method is also known as method of …………….

Select correct option:

**Successive displacement (Page 263)**
Iterations

False position

None of the given choices

Gauss elimination and Gauss-Jordan methods are popular among many methods for finding the ………of a matrix.

Select correct option:

Identity

Transpose

**Inverse**
None of the given choices

When the condition of diagonal dominance becomes true in Jacobi's Method.Then its means that the method is …………….

Select correct option:

Stable

Unstable

**Convergent**
Divergent

Under elimination methods, we consider, Gaussian elimination and ……………methods.

Select correct option:

Gauss-Seidel

Jacobi

**Gauss-Jordan elimination (Page 48)**
None of the given choices

The linear equation: 0x+0y =2 has -------- solution/solutions.

Select correct option:

**unique**
no solution

infinite many

finite many

If system of equations is inconsistent then its means that it has ………

Select correct option:

**No Solutions**
Many solutions

Infinite Many solutions

None of the given choices

In Jacobi's Method, We assume that the …………elements does not vanish.

Select correct option:

**Diagonal**

Off-diagonal

Row

Column

If the Relaxation method is applied on the system; 2x+3y = 1, 3x +2y = - 4, then largest residual in 1st iteration will reduce to ---------.

Select correct option:

**zero**
4

-1

-1

Full pivoting, in fact, is more ……………than the partial pivoting.

Select correct option:

Easiest

**Complicated**

While solving the system; x–2y = 1, x+4y = 4 by Gauss-Seidel method, which of the following ordering is feasible to have good approximate solution?

Select correct option:

x+4y = 1, x-2y = 4

x+2y = 1, x- 4y =4

x+4y = 4, x–2y = 1

**no need to reordering**

While solving by Gauss-Seidel method, which of the following is the first Iterative solution for the system; x-2y =1, x+4y=4 ?

Select correct option:

**(1, 0.75)**
(0,0)

(1,0)

(0,1)

If the determinant of a matrix A is not equal to zero then the system of equations will have……….

Select correct option:

**a unique solution**
many solutions

infinite many solutions

None of the given choices

The Gaussian elimination method fails if any one of the pivot elements becomes………

Select correct option:

One

**Zero**

While using Relaxation method, which of the following is the Residuals for 1st iteration on the system; 2x+3y = 1, 3x +2y =4 ?

Select correct option:

(2,3)

(3,-2)

(-2,3)

**(1,4)**

For the system; 2x+3y = 1, 3x +2y = - 4, if the iterative solution is (0,0) and 'dxi = 2' is the increment in 'y' then which of the following will be taken as next iterative solution?

Select correct option:

**(2,0)**
(0,3)

(0,2)

(1,-4)

Numerical methods for finding the solution of the system of equations are classified as direct and ………… methods

Select correct option:

Indirect

**Iterative**
Jacobi

None of the given choices

In Gauss-Seidel method, each equation of the system is solved for the unknown with -------- coefficient, in terms of remaining unknowns.

Select correct option:

smallest

largest

**any positive**

any negative

While using Relaxation method, which of the following is increment 'dxi'corresponding to the largest Residual for 1st iteration on the system; 2x+3y = 1, 3x +2y = - 4 ?

Select correct option:

-2

2

**3**
4

In full pivoting we interchange rows and columns such that the…………element in the matrix of the variables also get changed.

Select correct option:

Smallest

Middle

**Largest**
None of the given choices

Which of the following is the meaning of partial pivoting while employing the row transformations?

Select correct option:

**Making the largest element as pivot**
Making the smallest element as pivot

Making any element as pivot

Making zero elements as pivot

If a system of equations has a property that each of the equation possesses one large coefficient and the larger coefficients in the equations correspond to different unknowns in different equations, then which of the following iterative method id preferred to apply?

Select correct option:

Gauss-Seidel method

Gauss-Jordon method

**Gauss elimination method**
Crout's method

While using Relaxation method, which of the following is the largest Residual for 1st iteration on the system; 2x+3y = 1, 3x +2y = - 4 ?

Select correct option:

-4

3

**2**
1

For two matrices A and B, such that “A = Inverse of B”, then which of the following is true?

Select correct option:

B = Inverse of A

A and B are non-singular

AB = Identity Matrix

**All choices are true**

...........Remember for Pray me............

...........Best Of Luck.......